

5. Worked example — Busway 1A (critical, 5-wire, 200% neutral)

Hypothetical loads chosen to exercise every load type. Replace with your characterized load list; the method is the same for 1B, 2A, 2B, and 2C.

Step 1 — Calculate each load

ID	Load	Type	P (kW)	PF	Cont.	I (A)	S (kVA)	Q (kvar)
L1	Rack A PDU	1Ø 208	4.50	0.95	Y	22.77	4.737	1.479
L2	Rack B PDU	3Ø 208	12.00	0.95	Y	35.06	12.632	3.944
L3	Network rack	1Ø 120	1.80	0.98	Y	15.31	1.837	0.366
L4	Comms equipment	1Ø 120	1.20	0.90	Y	11.11	1.333	0.581
L5	Secure equipment	1Ø 208	3.00	0.92	Y	15.68	3.261	1.278
L6	In-row cooler (MCA 22 / MOP 35)	3Ø 208	5.11	0.85	Y	16.69	6.012	3.167
L7	Rack C PDU	1Ø 208	4.00	0.95	Y	20.24	4.211	1.315
L8	Maint. receptacles (6 × 180 VA)	1Ø 120	1.08	1.00	N	9.00	1.080	0.000

$$L1: 4500 / (208 \times 0.95) = 22.77 \text{ A}$$

$$L3: 1800 / (120 \times 0.98) = 15.31 \text{ A}$$

$$L2: 12,000 / (360.3 \times 0.95) = 35.06 \text{ A}$$

$$L6: 5110 / (360.3 \times 0.85) = 16.70 \text{ A}$$

Step 2 — Size branch breakers and outputs

ID	1.25 × I	Breaker	Output	Conductor
L1	28.5 A	2P 30 A	L6-30R	#10 Cu
L2	43.8 A	3P 50 A	CS8365C cord drop (368.56(B), Art. 400)	#6 Cu
L3	19.1 A	1P 20 A	L5-20R	#12 Cu
L4	13.9 A	1P 20 A	5-20R	#12 Cu
L5	19.6 A	2P 20 A	L6-20R	#12 Cu
L6	MCA 22 A	3P 35 A (≤ MOP)	Hardwired whip to unit disconnect (440.14)	#10 Cu
L7	25.3 A	2P 30 A	L6-30R	#10 Cu
L8	9.0 A (non-cont.)	1P 20 A	Whip to receptacle circuit	#12 Cu

Step 3 — Assign phases and balance

Order: 3Ø loads first → rotate 208 V 1Ø loads A-B, B-C, C-A → largest 120 V loads onto the lightest phases.

Contribution	A	B	C
L2 + L6 (3Ø): 35.06 + 16.70	51.76	51.76	51.76
L1 on A-B	22.77	22.77	—
L5 on B-C	—	15.68	15.68
L7 on C-A	20.24	—	20.24
L3 on C (lightest, largest 120 V)	—	—	15.31
L4 on B	—	11.11	—
L8 on A	9.00	—	—
Total (A)	103.76	101.31	102.98

$$I_{avg} = 102.68 \text{ A} \quad \text{Imbalance} = (103.76 - 102.68) / 102.68 = 1.1\% \quad \checkmark \leq 10\%$$

Step 4 — Busway and source checks

$$P_{\text{total}} = 32.69 \text{ kW} \quad Q_{\text{total}} = 12.13 \text{ kvar}$$

$$S_{\text{total}} = \sqrt{(32.69^2 + 12.13^2)} = 34.87 \text{ kVA} \quad \text{PF} = 0.94 \quad I_{\text{equiv}} = 96.8 \text{ A}$$

Check	Calculation	Result
PDR loading	$32.69 / 50 \text{ kW} = 65\%$	✓ ≤ 80%
Feeder / busway design current	$1.25 \times (103.76 - 9.00) + 9.00 = 127.5 \text{ A}$	✓ ≤ 175 A OCP, ≤ 250 A bus
Feeder conductor	175 A OCP → 2/0 Cu (175 A @ 75 °C); with 5 CCC: $195 \times 0.80 = 156 \text{ A} < 175 \text{ A}$	✗ upsize to 3/0 ($225 \times 0.8 = 180 \text{ A}$) or 4/0
Voltage drop, PDR → end feed (2/0, 60 ft)	$\sqrt{3} \times 12.9 \times 103.76 \times 60 / 133,100 = 1.05 \text{ V}$	0.50% ✓ (lower with 3/0)
EGC / IG	175 A OCP → #6 Cu each	✓

The 5-wire / 200% neutral choice changed the feeder: the two parallel neutrals make 5 current-carrying conductors, so 2/0 no longer carries a 175 A OCP. That is exactly the kind of thing the 5-wire decision triggers — recheck every feeder.

Step 5 — Neutral current with harmonics (why 200% neutral)

L-N loads: A = 9.00 A (L8, linear) B = 11.11 A (L4) C = 15.31 A (L3)

Fundamental: $I_{N,1} = \sqrt{(A^2+B^2+C^2 - AB - BC - CA)} = 5.6 \text{ A}$

L3, L4 switch-mode, $h3 = 0.80$:

$$I_{N,3} = 0.80 \times (11.11 + 15.31) = 21.1 \text{ A} \quad (\text{triplens add})$$

Total: $I_N = \sqrt{(5.6^2 + 21.1^2)} = 21.9 \text{ A}$

→ exceeds the largest L-N phase load (15.3 A)

Here the absolute neutral current is still small because most load is L-L or 3Ø. On a track dominated by 120 V IT loads, the same math drives the neutral toward $1.7 \times$ phase current — that is when the 200% neutral pays off.

Step 6 — Group into plug-in units (tap boxes)

Plug-in unit	Location	Circuits	Phases	Output	IG?
PIU-1A-01	Over rack A	L1 (2P30), L3 (1P20), L4 (1P20)	A-B, C, B	3 receptacles	Yes
PIU-1A-02	Over rack B	L2 (3P50)	A-B-C	Cord drop	Yes
PIU-1A-03	Over rack C	L7 (2P30), L5 (2P20)	C-A, B-C	2 receptacles	Yes
PIU-1A-04	Over cooler	L6 (3P35)	A-B-C	Hardwired whip	No (EGC only)
PIU-1A-05	Wall end	L8 (1P20)	A	Whip to receptacles	No
Spare	Reserved track	~20% capacity reserved	Lightest phase	—	—

Result: 5 plug-in units + 1 spare for this run.

6. Plug-in unit (tap box) procedure

- **Build the circuit list** from the 11 panel schedules: circuit ID, room, drop location, load, V, poles, PF, continuous?, critical?, nonlinear?, IG needed?
- **Assign each circuit to a busway** (1A/1B critical, 2A/2B/2C non-critical) by room and HEMP zone.
- **Group by drop location** — rack, equipment, receptacle cluster. One location = one plug-in unit.
- **Size each circuit** (Section 3.4 / 3.12) and pick the output: receptacle (match rating, Table 210.21(B)(3)), cord drop (368.56(B), 400.10 strain relief), or hardwired whip in flexible metal conduit (Art. 348) to a junction box or equipment disconnect.
- **Configure the unit:** breaker positions and poles, phase connection per breaker (fixed at the factory — record it), IG vs standard ground, metering, 100%-rated or standard. Split a location into two units if it

exceeds the positions per unit.

- **Balance** (Section 3.2) — iterate phase assignments until imbalance $\leq 10\%$ per track.
- **Check the track:** design current, PDR/transformer capacity, VD, neutral with harmonics.
- **Spares:** 10–20% spare units or reserved track length, with phase assignment chosen for the lightest phase.
- **Multiwire branch circuits** (shared neutral): common-trip or handle-tied breakers (210.4(B)) and grouping (210.4(D)). Separate neutrals avoid this.
- **Layout:** draw each track to scale with unit positions, drop lengths, supports (368.30), hook-stick access (240.24(A)(4), 404.8(A) Exc. 1).
- **Label** each unit: source busway, phase(s), circuit IDs, voltage; keep a busway schedule as the 408.4 directory equivalent (110.22(A)).